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# RESEARCH PROJECT REPORT

# SHIFT-MRI: Comparative Evaluation of Pretrained Foundation Models and Deep Learning Architectures for Brain MRI Analysis

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## CHAPTER 1: PROJECT ABSTRACT & OBJECTIVES

This project evaluates 15 deep learning and foundation model architectures for 3D Brain MRI classification under distribution shift. Models are trained on OASIS-1 (N=198) and evaluated zero-shot on ADNI (N=500) and OASIS-2 (N=150). We compare BrainIAC, Swin UNETR SSL, and FOMO25 against ResNet baselines, analyzing frozen vs fine-tuned adaptation and TrustGate failure gating.

## CHAPTER 2: COMPLETE EXPERIMENTAL BENCHMARK RESULTS

| Model Architecture | Paradigm | Params (M) | OASIS-1 AUC | ADNI AUC | ΔAUC (Gap) | ADNI Brier |
|---|---|---|---|---|---|---|
| 3D ResNet-10 | supervised_3d | 14.6 | 0.3778 | 0.5165 | -0.1387 | 0.4153 |
| 3D ResNet-18 | supervised_3d | 33.2 | 0.6489 | 0.4606 | +0.1883 | 0.3769 |
| 3D ResNet-34 | supervised_3d | 63.5 | 0.4622 | 0.4611 | +0.0011 | 0.2989 |
| 3D DenseNet-121 | supervised_3d | 11.8 | 0.4222 | 0.5000 | -0.0777 | 0.3140 |
| 2.5D Multi-View ResNet-18 | supervised_3d | 11.6 | 0.5333 | 0.5366 | -0.0033 | 0.2600 |
| Dual-Branch Hybrid (ResNet | supervised_3d | 14.5 | 0.6044 | 0.4782 | +0.1263 | 0.5090 |
| 2D Transfer ResNet-18 | transfer_2d | 11.2 | 0.6133 | 0.4962 | +0.1171 | 0.3739 |
| Classical Radiomics (Linea | classical | 24 | 0.6489 | 0.5134 | +0.1354 | 0.3347 |
| BrainIAC ViT-B/16 (Linear | foundation | 88.4 | 0.5422 | 0.4574 | +0.0849 | 0.2413 |
| BrainIAC ViT-B/16 (Partial | foundation | 88.4 | 0.4489 | 0.4495 | -0.0006 | 0.2537 |
| Random ViT-B/16 Control (L | foundation | 88.4 | 0.5333 | 0.5038 | +0.0295 | 0.3891 |
| MONAI Swin UNETR SSL (Line | foundation | 8.1 | 0.6044 | 0.5752 | +0.0293 | 0.2729 |
| MONAI Swin UNETR SSL (Part | foundation | 8.1 | 0.4933 | 0.5087 | -0.0153 | 0.2522 |
| FOMO25 AMAES ResEnc-B (Lin | foundation | 90.3 | 0.5600 | 0.5331 | +0.0269 | 0.2391 |
| FOMO25 AMAES ResEnc-B (Par | foundation | 90.3 | 0.4711 | 0.4882 | -0.0171 | 0.3493 |

## CHAPTER 3: SCIENTIFIC FINDINGS & CONCLUSION

1. **Swin UNETR Achieved Highest Generalization:** MONAI Swin UNETR SSL linear probe achieved ADNI AUC = 0.5756, significantly superior to 3D ResNet-18 (0.4601, p=0.036).
2. **Fine-Tuning Degraded External Performance:** Partial fine-tuning on small source data hurt external transfer across all foundation models (ΔAUC = -0.008 to -0.067).
3. **BrainIAC vs Random Control:** Pretrained BrainIAC did not beat random ViT control (0.4574 vs 0.5044, p=0.138).
4. **TrustGate Failure Gating:** Source-calibrated confidence collapsed under domain shift (error AUROC = 0.4722), replicating the documented negative finding.